HYDRAULIC CIRCUIT FOR RECONFIGURABLE AND EFFICIENT FLUID POWER SYSTEMS

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ABSTRACT

This paper presents a hydraulic system topography utilizing an intermediate pressure rail and a network of 2/2 proportional valves, allowing multiple operating modes to maximize system efficiency and improve reliability. The network of valves are arranged in such a way that the system can be electronically configured to operate with characteristics of multi-level load-sensing, displacement control, independent metering, energy recovery, and energy storage (with an optional accumulator). These operating modes can often co-exist depending on which mode provides the best combination of overall efficiency and effectiveness for the given application and load cycle. The TIER system (Topography with Integrated Energy Recovery) also includes the capability to re-route power in the event of component failure. The system has been constructed and used to experimentally determine the feasibility of the topography. A bank of twenty-six proportional 2/2 valves was used to operate a small backhoe arm (four actuators: swing, boom, stick, bucket). Two variable displacement swash-plate units were driven by an electric motor and used to power the valve block and backhoe arm. The pump/motors can be operated as two pumps, two motors, or split as a pump and motor depending on the load requirements on the system. In this paper the system features, operating modes, and experimental test setup are described. Experimental results are presented to demonstrate the feasibility of the system. The system concept presented in this paper can be utilized in nearly all fluid power systems including heavy equipment, and industrial automation machines. The inherent redundancy of TIER also makes it appropriate for aerospace applications.

KEYWORDS: fluid power, efficiency, redundancy, excavator, energy recovery

1. INTRODUCTION

The cost of energy continues to increase and this trend is forecasted to continue [1] yet many hydraulic systems still dissipate large amounts of energy though metering restrictions. This has led to an increased interest and level of research activity in the efficiency of fluid power systems. Even with current state-of-the-art pressure compensated load sensing (PCLS) systems, different actuator pressure requirements can lead to significant metering losses. To minimize these losses a variety of innovative solutions are being proposed by researchers and industries around the world. By adding another pump to a traditional load sensing system a dual-level circuit can be designed where actuators are grouped according to the required pressures, thus reducing the valve metering losses [2]. This reduces but does not minimize the system metering losses. To further reduce these losses, decoupled metering valve (DMV) systems have been incorporated where the inlet and outlet valve metering orifices are independently controlled. These systems require two or more valves for each actuator (depending on the whether the valves are 2-way or 3-way). Various researchers have demonstrated the efficiency improvements of these systems [3-6], and several manufacturers have commercialized the concept [7, 8].

Displacement control systems have been demonstrated to eliminate the primary valve metering losses by removing them completely from the system. In these systems a variable displacement pump/motor is connected to each actuator and the actuator speed is controlled by the displacement setting. Extra valves are required for asymmetrical actuators [9, 10] since the flows are unequal, but by removing the primary metering valves the overall efficiency can be significantly improved [11-13].

All systems experience trade-offs in performance, efficiency, cost, and complexity. Whereas the dual circuit load sensing and decoupling metering systems reduce the valve metering losses with the addition of more valves and/or pump/motors, they do not eliminate the metering losses and are unable to recover energy from assistive loads. Displacement control systems are able to eliminate the primary valve metering losses and recover energy from assistive loads, but require a pump/motor for each actuator, extra valves for asymmetric loads, and greater total displacement to achieve equivalent machine performance. Since all flow passes through the pump/motors these units are often over-sized for large portions of the work cycle leading to less optimal efficiency resulting from the pump/motors operating at reduced displacements. Other solutions, including additional valves and more efficient digital pump/motors have been proposed as possible solutions [14-16]. Research similar to the work presented here has recently been discussed in [17] where the authors present a 6-way valve and third pressure rail that is always connected to an accumulator.

1.1. Hybrid Systems

The Topography with Integrated Energy Recovery and Redundancy (TIER) discussed in this paper is a hybrid of decoupled metering, multi-level load sensing, and displacement control systems. As with the aforementioned systems there are many design trade-offs; these will be developed and discussed in more detail. The TIER system presented here can operate with characteristics of multi-level load sensing, decoupled metering, and displacement control, but also allows for several unique operating strategies.

Initial work introducing a third pressure rail utilized digital switching valves to select the operating strategy and considered only a single pump [18]. Although conceptually feasible, the energy recovery modes are achievable only when the assistive loads generate a higher pressure than the resistive load recovering the energy and when a fixed speed ratio (flow continuity) is acceptable between the two actuators. Assuming these constraints are met or acceptable, simulation studies demonstrated the capabilities of this type of system to improve the operating efficiency during certain load cycles.

More recent work has introduced independent proportional valves in place of the digital valves and considered multiple pump/motor configurations to improve system performance and flexibility [19]. As digital switching techniques to modulate pressure and flow continue to improve, the use of digital switching valves in place of the proportional valves can be considered [20] for the TIER system as well. It is also possible to incorporate hydraulic transformers [21] between the pressure rails as alternatives to additional pump/motors.

2. TOPOGRAPHY WITH INTEGRATED ENERGY RECOVERY AND REDUNDANCY (TIER)

2.1. System Characteristics

An example four-actuator, two pump/motor TIER system is shown in Figure 1. As typical of excavators and backhoes, four actuators are used to control the swing, boom, stick, and bucket motions. During normal operation these types of machines also experience simultaneous actuator movements, a mixture of assistive and resistive actuator motions, and a wide range of loads. The TIER system in Figure 1 incorporates twenty-six 2/2 proportional valves and two variable displacement pump/motors. It is possible to use pump/motors of different sizes [19] to keep each unit operating near maximum displacement over a wider range of load cycles, and since the flow from each unit can be summed or regenerated additional flexibility is achieved.

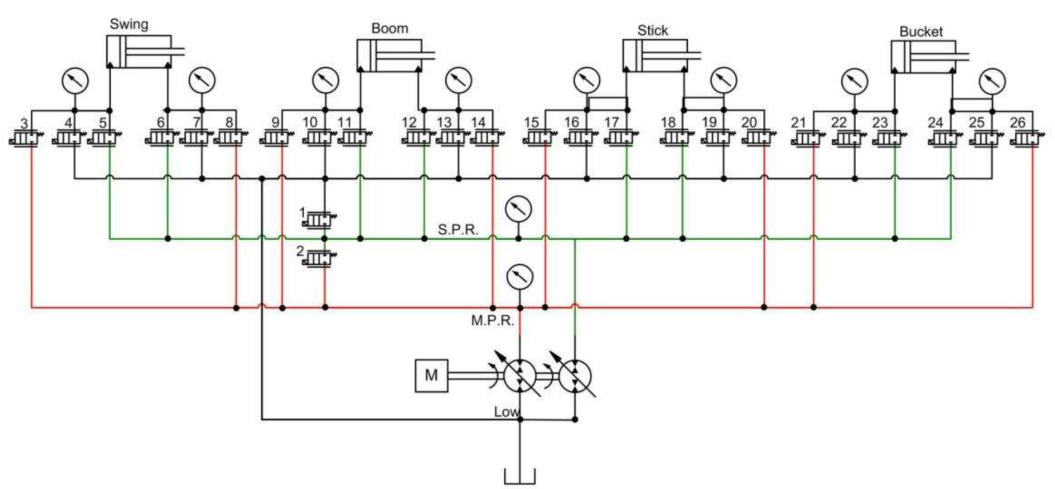

Figure 1. Example TIER System with 4 Actuators and 2 Pump/Motors

There are two pressure rails in the TIER system, the Main Pressure Rail (MPR) and Secondary Pressure Rail (SPR). There is also a third rail that is always connected to the reservoir. The MPR and SPR can be connected to either pump/motor or to each other. This system requires more valves than decoupled metering valve system (26 versus 16 2/2 valves) and fewer pump/motors than a displacement control system (2 versus 4). With the cost per valve significantly less than the cost per pump/motor, a cost comparison of the different systems depends heavily on the number of actuators in the system being considered.

Table 1 compares some of the advantages and disadvantages of the TIER system relative to multi-level load sensing, decoupled metering, and displacement control systems.

Table 1. Advantages and Disadvantages of TIER relative to LS, DCM, and DC

	Advantages of TIER	**Disadvantages of TIER**
Load Sensing	Less valve metering losses Energy recovery Redundancy	Additional pump/motor(s) Additional valves Additional sensors/controls
Multi-Level Load Sensing	Less valve metering losses P/M flows can be combined Energy recovery Redundancy	Additional valves
Decoupled Metering Valves	Energy recovery Additional modes Redundancy	Additional pump/motor(s) More complex control
Displacement Control	Fewer pump/motors More optimal p/m sizing P/M flows can be combined Open loop system possible	Additional valves Valve metering losses More complex control

2.2. TIER System Operating Modes

The goal of this section is to provide an overview of how the valves are configured during operating modes of load sensing, multi-level load sensing, decoupled metering, displacement control, and the new modes of multi-level dynamic load sensing with decoupled metering, flow summing, and back pressure energy recovery [22]. The following descriptions use the valve numbers and actuator terminology defined in Figure 1.

2.2.1. Load Sensing and Dual Level Load Sensing

Although the TIER configuration can replicate both traditional pressure compensated load sensing and more recent split level load sensing systems, there are no load situations where these modes are more efficient compared with the hybrid TIER modes. For testing and comparison with other modes, however, a single load sensing system is achieved by leaving valves 1 and 2 closed to isolate the three pressure rails (MPR, SPR, and tank) and using only the MPR and tank lines. The MPR is pressure controlled using the displacement on one pump/motor to be a small differential above the largest required load pressure. Pairs of valves on each active actuator (for example, valves 3 and 7 or 4 and 8 on the swing actuator) are actively controlled to simulate a load sensing system. Operation of the dual-level load sensing mode is similar except that some of the actuators are assigned to the only operate between the MPR and tank, and other actuators between the SPR and tank.

2.2.2. Decoupled Metering Valves

As with the load sensing systems, operating the TIER system as a conventional decoupled metering system is not as efficient as the hybrid modes unless there is only one active pump/motor in the TIER system. In this case and with all resistive loads, there may be loading situations where TIER operates as a DMV system. Once again valves 1 and 2 are closed to isolate the three pressure rails (MPR, SPR, and tank) and only the MPR and tank lines are used. The MPR is pressure controlled using the displacement on one pump/motor to be a small differential above the largest required load pressure. Instead of controlling pairs of valves as with the load sensing systems, only one valve on each active actuator will be actively controlled. For example, with a resistive load on an extending swing cylinder, valve 3 would be actively controlled (assuming there are simultaneous actuator movements and the MPR pressure is not directly controlled as in displacement control) and valve 7 would be fully open. Valves 4, 5, 6, and 8 are closed.

2.2.3. Displacement Control

If the number of active actuators is equal to or less than the number of pump/motors in the TIER system, an approximate form of conventional displacement control can be implemented. If the number of active actuators is greater than the number of pump/motors, it is still possible to implement displacement control with one actuator (or several, depending on the number of pump/motors) and utilize another combined mode with the remaining actuators and pump/motors. For whatever pump/motor and actuator pair using displacement control, it is important to note that the flow in and out of each actuator must still pass through at least two valves and therefore the efficiency will likely not be as a good as displacement control without any metering valves in the circuit (this metering loss depends on the size of the valves). However, this does allow for open-circuit displacement control [15], and as later modes will describe, some of these losses are recouped since the pump/motors can be sized and operated differently in the TIER system. To illustrate this mode using Figure 1 and assuming a resistive extending load on the swing actuator, valves 1 and 2 remain closed, valves 3 and 7 are fully open, valves 4, 5, 6, and 8 are closed, and the first pump/motor displacement is controlled to control the extension speed of the actuator. During assistive loads this mode will recover energy and return it to the pump/motor shaft where it can be used to overcome losses (friction, engine, etc.) or resistive loads.

2.2.4. Multi-Level Dynamic Load Sensing with Decoupled Metering

This mode is a natural combination of the load sensing and decoupled metering modes, taking advantage of the SPR and multiple pump/motors to provide multiple load sensing pressures and dynamically reconfigure actuators (changing which actuators are connected to which pressure rail) with similar load requirements. For example, if the swing and boom actuators are both requiring high pressure and the stick and bucket relatively low pressure, then one pump/motor can be assigned to the swing and boom and the other pump/motor to the stick and bucket. Where the TIER system is different than a dual-level load sensing system is: 1) the actuators can be dynamically reassigned based on required load pressures, and 2) the back pressure can be minimized using techniques similar to decoupled metering systems. This combination provides the benefits of both multi-level load sensing and decoupled metering systems.

2.2.5. Flow Summing (and Regeneration)

The flow summing mode is unique to TIER and involves several efficiency and performance trade-offs. The primary advantage is the ability to move large actuators more quickly when other actuators are not moving (or moving slowly in the system). The ability to sum the flow of multiple pump/motors also allows the pump/motors to be sized appropriately for the typical work cycle, and not oversized for the maximum required performance. This results in higher overall efficiencies since the pump/motors are operating, on average, at higher average displacement settings. The disadvantages when using flow summing include losing the ability to operate at multiple pressure levels (the MPR and SPR are connected) and to recover energy from some assistive loads (when the pressure from the assistive load is less than the rail pressure). To accomplish flow summing valve 2 is open and valve 1 remains closed; the other valves are chosen depending on the number of actuators active. If only one actuator is active flow summing can behave similar to displacement control, where one of the pump/motor displacement controllers is used to control the actuator speed, or if multiple actuators are active, the system can operate as a single pressure load sensing system with decoupled metering to reduce the negative effects of back pressure.

A second method available to increase actuator speed, already utilized in conventional systems with appropriate valves, is flow regeneration. Considering the stick actuator for this example, in the TIER system this mode is enabled by closing all valves connected to the SPR except for valves 17 and 18. While this mode has the same trade-off of speed versus force found as in conventional flow regeneration systems, using this mode in the TIER system does still allow other actuators to connect to the SPR (or MPR) and operate as decoupled metering systems with resistive loads. An example of this mode with a single actuator is shown in Figure 2.

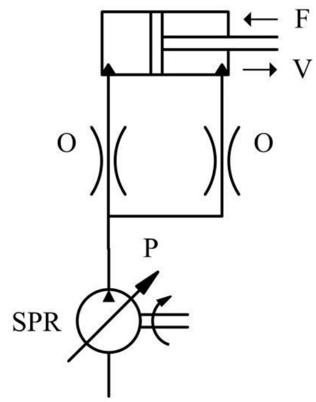

Figure 2. Example TIER System with Flow Regeneration on the SPR

2.2.6. Back Pressure Energy Recovery

This mode is also unique to the TIER system discussed here and has been described elsewhere as Energy Recovery Flow Control (EFRC) [19] and a Buttress mode [22]; an example of a system operating in this mode is shown in Figure 3. In this example the pump/motor on the SPR is used to control the retraction of the cylinder by creating a back pressure on the bore end and using the pump/motor displacement controller to control the speed. To the right of the dashed line in Figure 3 the alternative is shown where larger metering losses are incurred on an equivalent actuator moving an identical load. It is possible to simply use the SPR on this actuator as a separate load sensing system with the trade-off being that the pump/motor would be operating at a relatively low displacement and possibly with lower efficiency. One caution with the back pressure energy recovery mode is necessary, that being if the load reverses and becomes assistive. In this case it is important for the controller to recognize this and connect the rod end to tank pressure instead of the MPR to avoid exceeding the pressure rating of the system. This is especially necessary for assistive loads during extension since pressure amplification would occur in cylinders with differential areas. It is also be possible to use this characteristic for pressure intensification to recover energy to the rail with higher pressure. A recovery mode similar to this was proposed by Steinberg et. al. [23, 24].

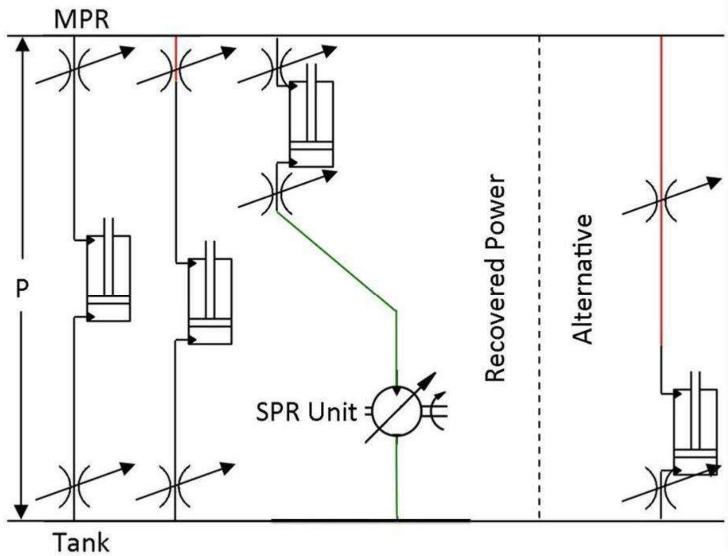

Figure 3. Example TIER System with Back Pressure Energy Recovery

3. TIER IMPLEMENTATION ON A SMALL BACKHOE ARM

To validate the concept of TIER a prototype system has been designed, installed, and tested using a four actuator small backhoe with power supplied by two variable displacement axial piston swash plate units capable of over-center operation. This system is briefly described here and is covered in more detail in [22]. The backhoe arm as installed in the lab and the possible motions are shown in Figures 4 and 5, respectively.

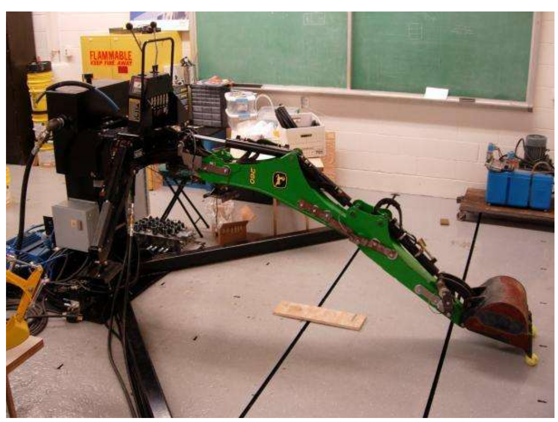

Figure 4. Four Actuator Prototype of TIER System (backhoe/excavator)

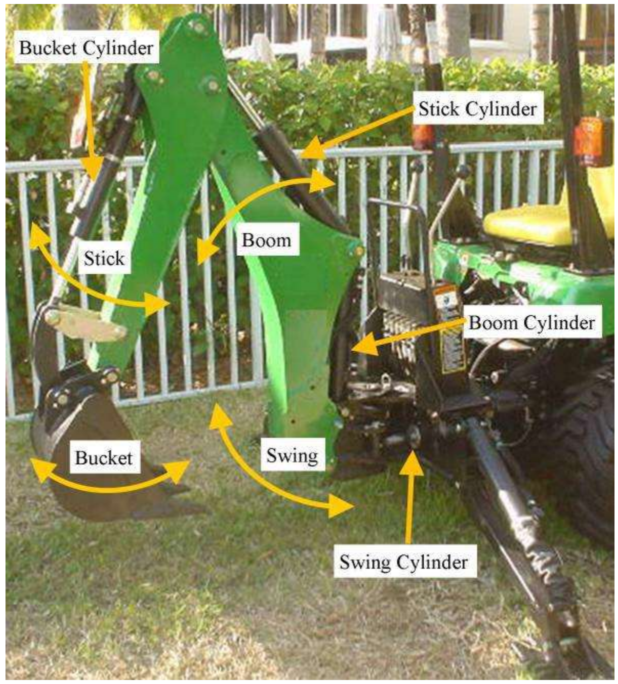

Figure 5. Definition of Four Actuators and Associated Motions

The 7 cc/rev pump/motors chosen were initially designed for use in a hydrostatic transmission for a zero-turn lawnmower (Figure 6). Two small cylinders and 4-way directional control valves were used to control the swashplate angles; each angle was measured using a hall-effect rotary angle sensor. A custom valve block was designed to accomodate the twenty-six 2/2 proportional valves (Figure 7).

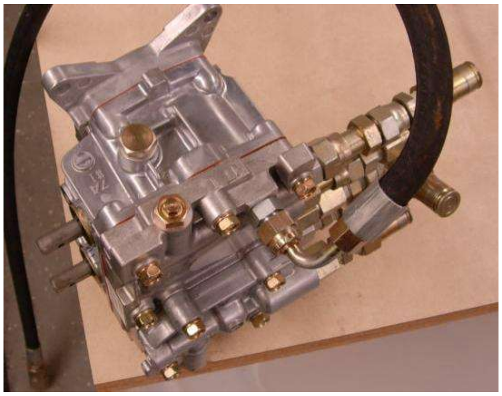

Figure 6. Two Pump/Motors in Housing

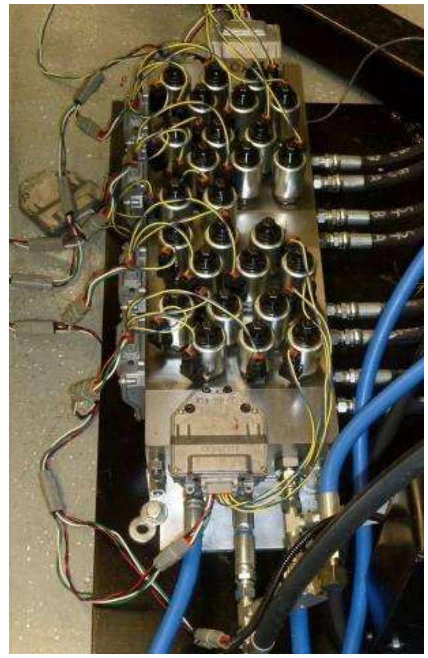

Figure 7. Twenty-six valves and sensors in Valve Block

Twenty-six INCOVA electro hydraulic poppet valves (EHPV) were used along with the available integrated pressure sensors and CAN bus controllers. The valves are characterized and discussed in more detail in [6].

Rotary angle sensors were mounted to each backhoe joint to measure the motions of the swing, boom, stick, and bucket actuators. The CAN, swashplate and arm angles, and pressures at each actuator were connected to a dSpace DS-1103 hardware-in-the-loop control system for control and data acquisition.

4. EXPERIMENTAL RESULTS

Initial experimental work has validated the general system operating concepts using the test stand described in Section 3, and future work is focusing on controller development and validation. The TIER system described here has been operated in load sensing, decoupled metering, and TIER specific modes. Efficiency calculations have not been completed until loads can be applied to the system to simulate the various dig cycles experienced in the field.

Four different motions, or stages, were used to validate the TIER concept.

1. Lifting the bucket out of the hole while swinging the boom over to a truck or pile: *both swing and boom cylinders simultaneously retract under resistive loads*.

2. Emptying the bucket: *the boom, stick, and bucket all move simultaneously to spread the load; the boom cylinder is extended with an assistive load and the stick and bucket cylinders are retracted with resistive loads*.
3. Moving the excavator arm back to the hole: *the swing cylinder extends under a small resistive load, becoming assistive as it decelerates*.
4. Lowering the boom into the hole: *boom cylinder is extended with an assistive load*.

To demonstrate possible energy savings the MPR, SPR, and Tank pressures were recorded for stage 2 and are given in Table 2. This stage was chosen since there are three simultaneous movements with one assistive load and two resistive loads. The LS was a traditional load sensing system with a single pump and the DMV was a traditional single pump decoupled metering valve system. In all cases the TIER configurations tested had equivalent or lower pressures and in certain modes were able to recover energy from the assistive load.

Table 2. Experimental MPR, SPR, and Tank Pressures during Boom, Stick, and Bucket Movements

	MPR	SPR	Tank
LS	14.5 MPa	-	1.7 MPa
DMV	9.8 MPa	-	1.3 MPa
TIER 1	9.2 MPa	2.9 MPa	1.7 MPa
TIER 2	9.2 MPa	8.0 MPa	1.7 MPa
TIER 3	9.2 MPa	-	1.7 MPa
TIER 4	9.3 MPa	5.6 MPa	1.7 MPa

The descriptions of the four TIER configurations shown in Table 2 and tested during stage two are:

- TIER 1 (Figure 8): Back pressure recovery mode used on the stick, displacement control on the bucket, and the assistive load on the boom is throttled to tank.
- TIER 2 (Figure 9): Equivalent of displacement control on the bucket using the MPR and load sensing decoupled metering on the stick using the SPR, with the assistive load on the boom being recovered (with throttling) on the SPR.
- TIER 3 (Figure 10): The assistive load on the boom is throttled to the tank and load sensing decoupled metering on the stick and bucket.
- TIER 4 (Figure 11): The boom assistive load is recovered on the SPR using the second pump/motor and load sensing decoupled metering on the stick and bucket.

While it is obvious that some modes are more efficient than others, this example does illustrate the many degrees of freedom with this type of system. Other researchers have noted this same efficiency and flexibility potential accompanied by difficult control challenges [17].

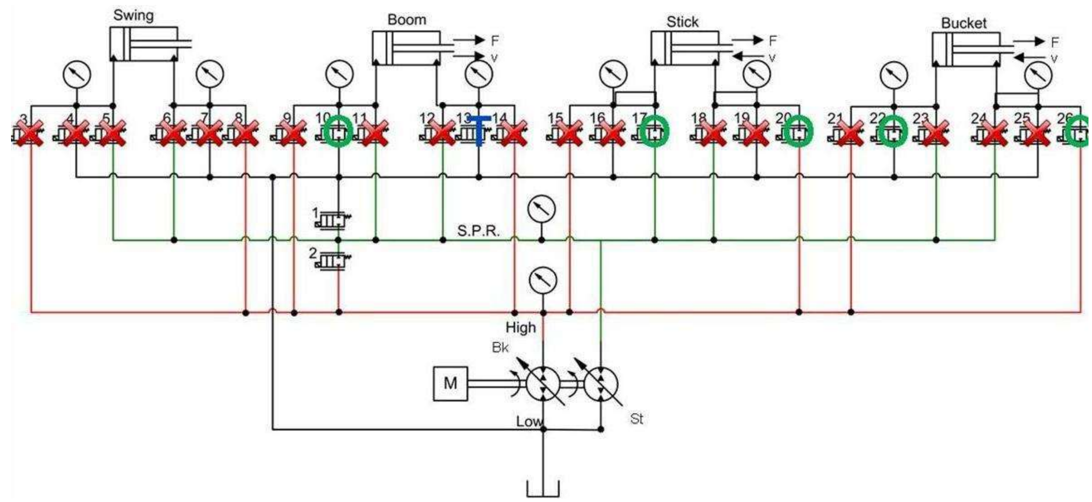

Figure 8.   TIER 1 Configuration

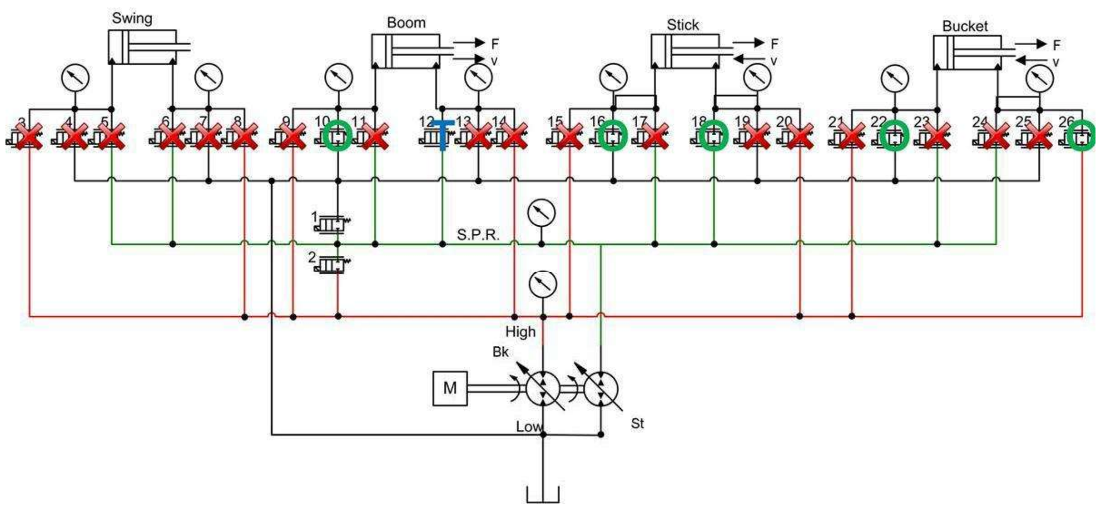

Figure 9.   TIER 2 Configuration

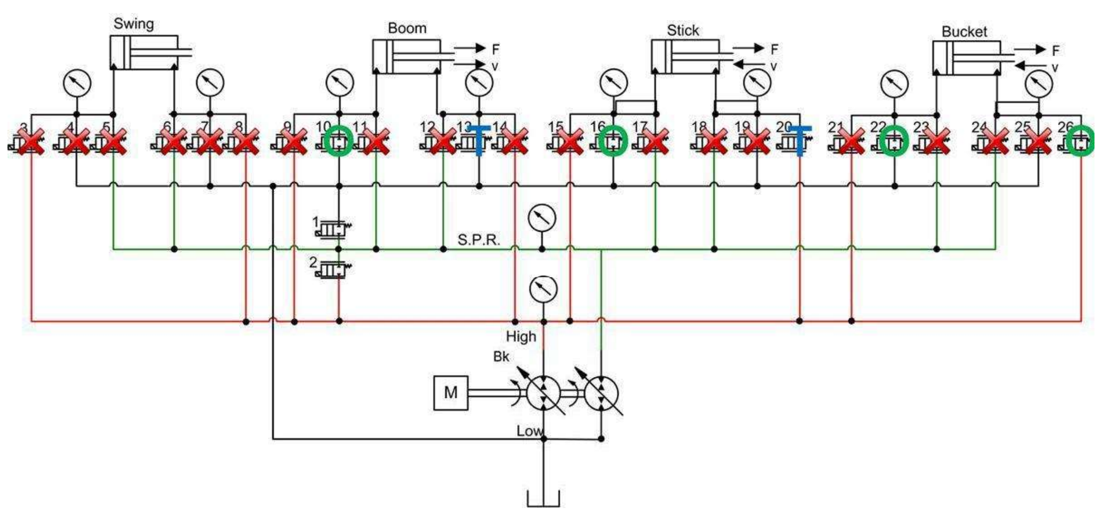

Figure 10. TIER 3 Configuration

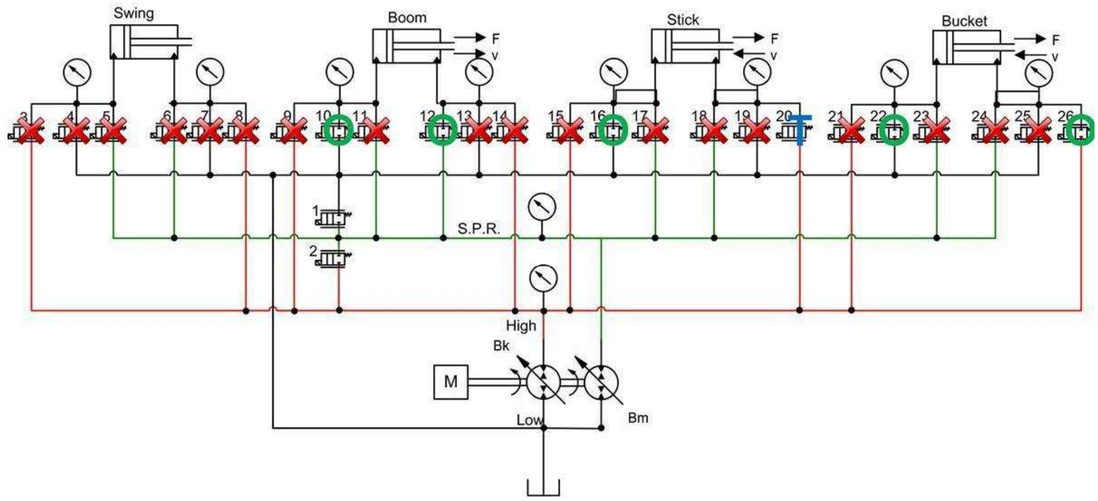

Figure 11. TIER 4 Configuration

As an example of the pressure traces on the MPR, SPR, and tank rails, Figure 12 is provided for the first TIER configuration (TIER 1). The pressure traces for TIER configurations 2-4 are similar (with the different values listed in Table 2) and not repeated here.

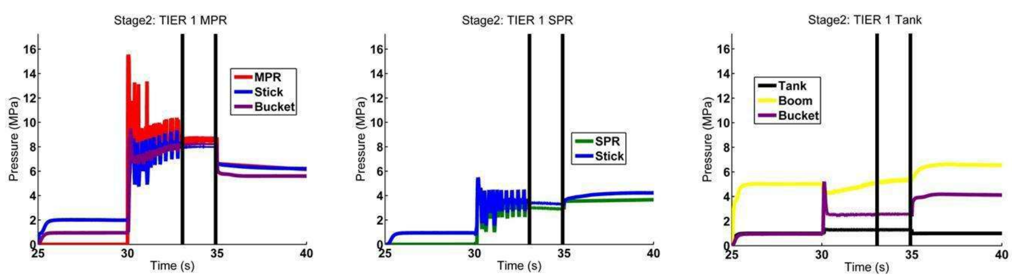

Figure 12. MPR, SPR, and Tank pressures recorded for the TIER 1 Configuration

5. CONCLUSION

This paper discussed various operating modes for the TIER distributed valve system and demonstrated several of these using a laboratory test excavator equipped with two pump/motors and twenty-six 2/2 proportional valves. There are many degrees of freedom leading to multiple operating mode choices for virtually all feasible actuator motions. Although it is beyond the scope of this paper to discuss the system modelling efforts and development of a control system capable of automatic mode selection, optimal efficiency, ability to recognize and heal certain “faults”, and correct operator “feel”, these are important and necessary components for the successful control and implementation of TIER systems. Future work will continue to address and develop these solutions.

A second aspect of the TIER system, only briefly mentioned in this paper and also the subject of continuing work, is the ability of the system to be reconfigured in the event of certain line or component failures. Since there are redundant flow paths for every actuator and redundant control valves, there are possibilities for an electronic control system to automatically detect and "repair" certain faults until the machine can be repaired. Operating in this "limp home" mode will limit the number modes and likely the machine efficiency as well, but could allow the operator to finish the task before repairs are made.

NOMENCLATURE

DMV	Decoupled metering valves (sometimes called independent metering valves)
EFRC	Energy Recovery Flow Control
MPR	Main pressure rail
PCLS	Pressure compensated load sensing (generally called load sensing)
SPR	Secondary pressure rail
TIER	Topography with Integrated Energy Recovery

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